

AAA Local Authentication

Identity Services Lab 2

A hands-on Cisco Packet Tracer security implementation

PROJECT HIGHLIGHTS

- Configured Cisco AAA with a named local authentication method list
- Applied AAA authentication to SSH VTY access on R1 and SW1
- Tested primary, backup, and unauthorized user identities
- Verified active sessions, VTY controls, and saved configurations

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Network Connectivity Before AAA Testing

VERIFIED

```
Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=3ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.2: bytes=32 time<1ms TTL=255
Reply from 192.168.10.2: bytes=32 time<1ms TTL=255
Reply from 192.168.10.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PROJECT EVIDENCE

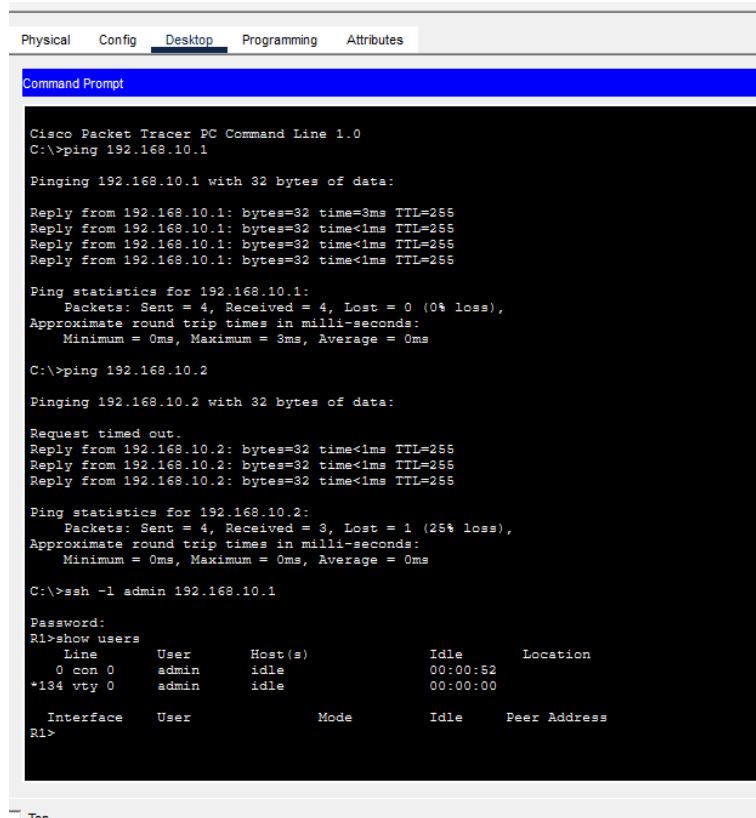
Confirmed that the administrator workstation could reach both managed devices before applying AAA. R1 returned four replies with zero packet loss. SW1 returned three replies after the first ARP-related timeout.

WHAT THIS DEMONSTRATES

- IPv4 connectivity verification
- Router and switch reachability
- Baseline testing before security changes
- Recognition of normal ARP behavior

Successful R1 AAA Local Login

VERIFIED



```
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=3ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.2: bytes=32 time<1ms TTL=255
Reply from 192.168.10.2: bytes=32 time<1ms TTL=255
Reply from 192.168.10.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ssh -l admin 192.168.10.1

Password:
R1>show users
  Line          User             Host(s)          Idle           Location
  *--*--*--*--*
    0 con 0      admin            idle             00:00:52
  *134 vty 0     admin            idle             00:00:00

  Interface    User             Mode             Idle           Peer Address
R1>
```

PROJECT EVIDENCE

Authenticated to R1 through SSH using the local administrator identity. The show users command confirmed an active VTY session for the admin account.

WHAT THIS DEMONSTRATES

- AAA-backed local identity authentication
- Secure SSH remote administration
- Active session verification
- Cisco IOS user access control

AAA Method List and VTY Controls

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```
R1#show running-config | section aaa
aaa new-model
aaa authentication login AAA-LOCAL local
R1#show running-config | section line vty
line vty 0 4
  exec-timeout 5 0
  login authentication AAA-LOCAL
  transport input ssh
R1#
```

VERIFIED CONFIGURATION

aaa new-model | aaa authentication login AAA-LOCAL local | login authentication AAA-LOCAL | transport input ssh

SKILLS: AAA method lists, VTY line security, SSH-only access, configuration validation

Unauthorized Logins Rejected on R1 and SW1

BLOCKED

R1 Unauthorized Login Test

```
Interface User
R1>ssh -l unauthorized 192.168.10.1
Password:
% Login invalid
Password:
```

The nonexistent username was rejected, proving that R1 accepts only identities stored in the configured local database.

SW1 Unauthorized Login Test

```
C:\>ssh -l unauthorized 192.168.10.2
Password:
% Login invalid
Password:
Password:
```

The same negative test was repeated against SW1, confirming consistent access-control behavior across both devices.

Technical Outcomes and Skills

VERIFIED

CONFIGURATION OUTCOMES

- Enabled Cisco AAA using aaa new-model
- Created the AAA-LOCAL authentication method list
- Protected VTY access with local AAA and SSH
- Verified valid and invalid identity attempts

SKILLS DEMONSTRATED

- Cisco Packet Tracer and Cisco IOS
- Authentication, Authorization, and Accounting
- Local identity management and SSH security
- Access-control testing and troubleshooting

PROJECT OUTCOME

This project demonstrates the ability to configure AAA local authentication on Cisco network devices, apply the method list to SSH VTY lines, confirm active administrator sessions, and reject unauthorized identities. During troubleshooting, Packet Tracer's limited AAA EXEC authorization support required privileged access to remain protected by a separate enable secret. The final configuration was verified and saved on both R1 and SW1.

PORTFOLIO-READY DELIVERABLE: Built, tested, troubleshot, documented, and prepared for LinkedIn media.